MOHAN RAJ SENTHILNATHAN

PRINCIPAL SOFTWARE ENGINEER
DISTRIBUTED SYSTEMS | AI AGENTS

mohansenthil92@gmail.com
+91 90431 21430
Chennai, India
linkedin.com/in/mohanras

Principal Software Engineer with 12+ years of experience building secure, cloud-native distributed systems, including a multi-tenant platform serving 400K daily active users across Norton, Avast, and Avira. Currently building production AI agents (multi-agent orchestration, RAG, Claude Agent SDK) that automate defect creation, triage, and remediation, while mentoring engineers and driving AI adoption across the team.

Technical Skills

- Languages: C#, Java, Python, JavaScript, TypeScript
- AI & Agents**:** Claude Agent SDK, Claude API, AWS Bedrock, LangGraph, Multi-Agent Orchestration, MCP, Agent Skills, RAG (Pinecone hybrid search), Tool Calling, Prompt Engineering, Langfuse
- Frameworks: .NET Core, Spring Boot, Django, React, Angular, Node.js
- Data & Messaging: PostgreSQL, Cosmos DB, Redis, BigQuery, Pinecone, Kafka, RabbitMQ, Pub/Sub
- Cloud & DevOps: Azure, GCP, AWS, Kubernetes, Docker, Nginx, TeamCity, Azure DevOps, Git, CI/CD
- Architecture: Microservices, Event-Driven, Multi-Tenancy, CQRS, Domain-Driven Design, Circuit Breaker / Resilience Patterns, Database Sharding, REST APIs
- Observability: Prometheus, Grafana, Datadog, Splunk
- Security: OAuth 2.0, OIDC, JWT, OWASP Practices

Highlights

- **Scale:** Architected a cloud-native multi-tenant platform serving 400K daily active users across Norton, Avast, and Avira
- **AI for Engineering Productivity:** Built a RAG defect-creation tool (Pinecone hybrid search) that raises 80% of defects, plus multi-agent remediation on Claude Agent SDK that resolves up to 60% of regression failures autonomously.
- **AI for Customer Experience:** Built a LangGraph multi-agent service that detects IMAP settings and generates localized setup guides, making email onboarding self-service and easy to extend to new providers.
- **Full-Stack + Distributed Systems:** Deep experience across .NET/C#, Java/Spring Boot, React, Kafka, RabbitMQ, Redis, Cosmos DB, and PostgreSQL on Azure and GCP
- **Reliability & Security:** Designed circuit breaker failover, OIDC-based authentication, and database sharding for multi-tenant production systems
- **Technical Leadership:** Mentor engineers, lead code reviews and epic estimation, and drive team-wide AI tool adoption

Professional Experience

Gen (NortonLifeLock / Symantec) – Chennai (9+ Years)

- Principal Software Engineer | Jul 2023 – Present
- Senior Software Engineer | Aug 2020 – Jul 2023
- Software Engineer | Jan 2017 – Aug 2020

Key Contributions in MultiTenant Web Portal (Since Jan 2017):

- **Architected and delivered a cloud-native multi-tenant platform** using .NET Core, React, Azure, Redis, and Cosmos DB; scaled across Norton, Avast, and Avira serving 400K daily active users.
- **Built a test result classification system** with the Claude Agent SDK that triages CI/CD failures, cutting the manual analysis needed to spot real defects. (AI)
- **Built a RAG-based AI defect-creation tool** that turns free-text bug reports into structured Jira tickets. It retrieves context from a curated product-flow knowledge base using Pinecone hybrid search (dense + sparse) to identify the right use case, then generates tickets via schema-validated tool calling, with Langfuse tracing for observability. Now used to raise 80% of defects. (AI)
- **Built a two-agent remediation framework** that fixes up to 60% of regression failures with no human involvement. The first agent pulls logs through MCP and skills, finds the root cause, and comments when it's an application issue. The second takes that root cause and implements the fix. (AI**)**
- **Used Claude Agent SDK** to analyze sprint check-ins and flag regression hotspots, so testing could focus on the riskiest areas. (AI)
- **Designed a circuit breaker with automatic failover** for critical backend systems in a multi-tenant customer portal. It trips on a configurable failure ratio over a sampling window and counts only true connectivity failures
- **Designed the OIDC-based authentication and authorization** layer for the Gen portal, with RESTful APIs shared across the multi-tenant services.
- **Cut Cosmos DB RU consumption** by querying only the fields needed and adding custom indexes.
- **Added a Redis cache** to keep data in sync across servers and reduce backend calls**.**

Technologies used in MultiTenant Web Portal:

.NET, C#, JavaScript, React, Python, Azure, Cosmos DB, Redis, Claude Agent SDK, Pinecone, TeamCity

Key Contributions in Email Guardian (Since Oct 2024):

- **Built a multi-agent service on LangGraph** that detects a user's IMAP settings and generates localized email setup guides. Onboarding is now self-service, support effort dropped, and adding a new provider is easy. (AI)
- **Designed the Feedback Loop** that lets customers report incorrectly scanned emails and triggers an automatic re-scan. Built the full **data flow on Pub/Sub**.
- **Designed a Metrics API on Kafka** and GCP BigQuery that collects data in real time and scales with traffic
- Built an internal web app for customer support so engineers can look up and resolve queries about onboarded emails.
- **Built a microservice orchestrator** on RabbitMQ to run backend workflows through reliable, message-driven communication.
- **Replaced paid Nginx** authentication features with open-source modules, which let us drop the licensed tier.
- **Implemented database sharding** for horizontal scalability, which allowed the Norton tenant to be onboarded onto the platform.

Technologies used in Email Guardian:

Java, Spring Boot, Kubernetes, Docker, Kafka, PostgreSQL, Nginx, GCP, Team city, Grafana, Python.

Other Contributions in Current Company (Leadership Role):

- **Mentored junior engineers** and ran knowledge-sharing sessions, technical workshops, and design discussions for the team.
- **Led epic estimation** and dependency mapping across teams and raised technical blockers early.
- **Led code reviews** and pushed the team toward consistent development practices.
- **Drove AI tool adoption in the team**, covering prompt engineering, rules, and spec design for AI-assisted development. (AI)
- **Built open-source** C# and JS libraries for reuse across projects.

HCL Technologies (Client: Cisco Systems) – Chennai

- Member Technical Staff | Mar 2015 – Dec 2016

Key Contributions in Cisco Identity Services Engine:

• Developed REST APIs and backend services for Cisco Identity Services Engine (ISE) platform.
• Built internal developer productivity tools improving turnaround time for development and QA teams.

Amazon Development Center – Chennai

- Device Associate | Oct 2013 – Mar 2015

Key Contributions in Project B (Amazon Fire Phone):

• Owned module-level testing for Amazon Fire Phone & Kindle devices, automating key test cases to enhance release quality.

Education

B. Tech in Information Technology | Anna University | 2009 – 2013

Certifications

Microsoft Certified: Azure Fundamentals (AZ-900) – 2022

Patent

Systems and Methods for Restricting Capture of Self-Portrait Images on Mobile Devices
Patent No: 11528355 | Issued: Dec 13, 2022